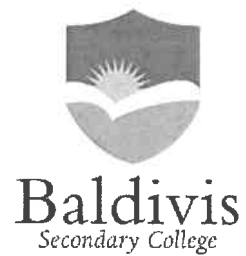


Year 10
Pathway 10A



Term 3 Test
Statistics and Linear Relationships
Section One

Name: Suluhans

Score: /46
Total Time: 45 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks. Calculators may be used One A4 notes (handwritten) allowed.

Question 1

(12 marks)

Solve the following equations

a) $7x + 5 = 47$

$$7x = 42$$

$$x = 6 \quad \checkmark$$

b) $3p - 2 = -7$

$$3p = -5$$

$$p = -\frac{5}{3} \quad \checkmark$$

c) $\frac{t+5}{2} = 1$

$$t+5 = 2$$

$$t = -3 \quad \checkmark$$

d) $5(2y + 3) = 15$

$$2y+3 = 3$$

$$2y = 0$$

$$y = 0 \quad \checkmark$$

e) $4x + 3 = 9x - 2$

$$5x = 5$$

$$x = 1 \quad \checkmark$$

f) $12 - x = 3(x - 5)$

$$12 - x = 3x - 15$$

$$4x = 27$$

$$x = \frac{27}{4}$$

$$= 6.75 \quad \checkmark$$

Question 2

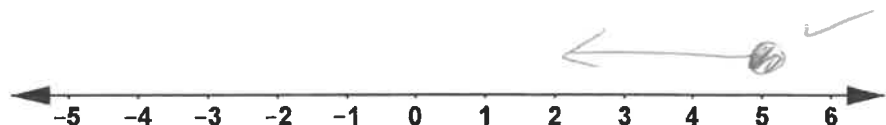
(4 marks)

Solve the following in equalities and represent the solution on a number line

a) $2x - 1 \leq 9$

$$2x \leq 10$$

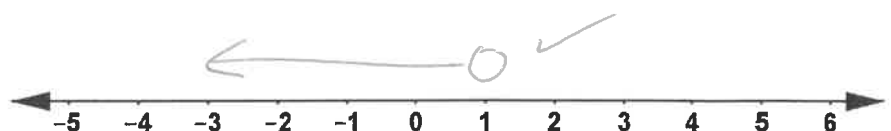
$$x \leq 5 \quad \checkmark$$



b) $8 - 5x > 3$

$$-5x > -5$$

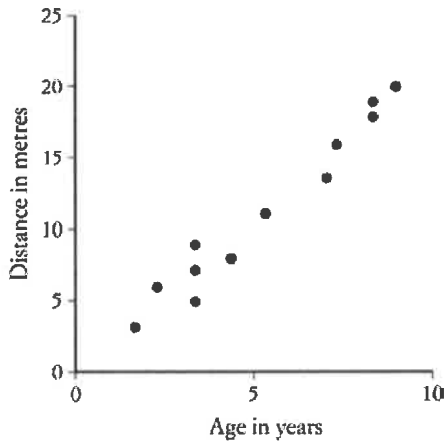
$$x < 1 \quad \checkmark$$



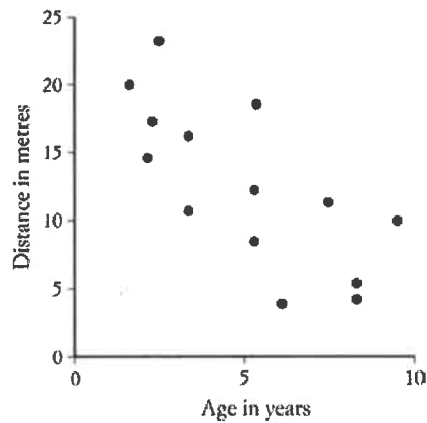
Question 3

(4 marks)

Describe the relationship between the variables in the scatter plots below.



Strong positive
correlation.



Weak negative

Question 4

(4 marks)

8 9 9 12 10 13 14 8 4 5 7 10 7 9 8 7 5 7

For the data set find

a) the interquartile range for the data?

$$10 - 7 = 3$$

b) the mean

$$8.4$$

c) standard deviation

$$2.61$$

Question 7

(4 marks)

Look at the 5 equations below

A	B	C	D	E
$y = 4x + 1$	$4x + y = 1$ $y = -4x + 1$	$4x - y = 1$ $y = 4x - 1$	$4x + 4y = 1$ $y = -x + \frac{1}{4}$	$4y = x + 1$ $y = \frac{1}{4}x + \frac{1}{4}$

a) Which two lines are parallel? Why?

A & C Same gradient 4.

b) Which two lines are perpendicular? Why?

B and E
 $-4 \times \frac{1}{4} = -1$ Gradient multiply to give -1

Question 8

(2 marks)

Find the equation of a line perpendicular to the line $y = -3x + 2$ that has a y-intercept of 5.

perpendicular $y \neq m = \frac{1}{3}$

$$y = \frac{1}{3}x + 5$$

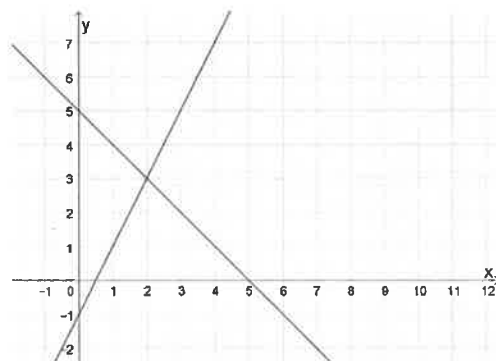
Question 5

(2 marks)

Use the graph below to find the solution to the pair of simultaneous equations.

$$\begin{cases} x + y = 5 \\ y = 2x - 1 \end{cases}$$

$$\begin{aligned} x &= 2 \\ y &= 3 \end{aligned}$$

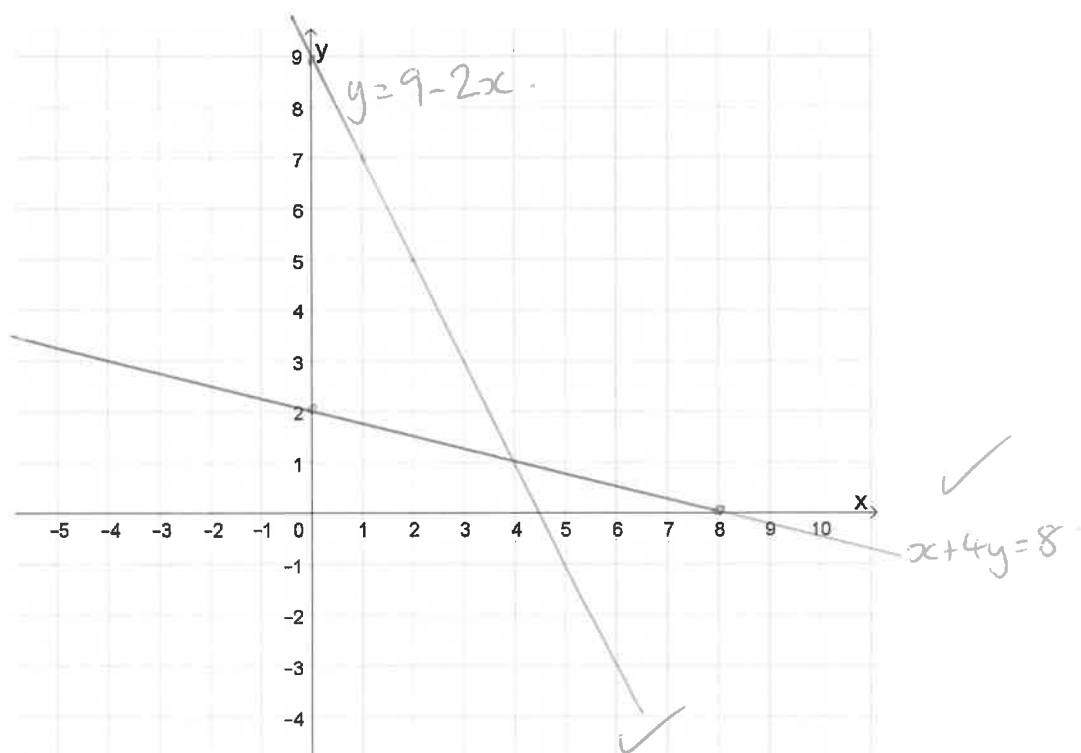


Question 6

(4 marks)

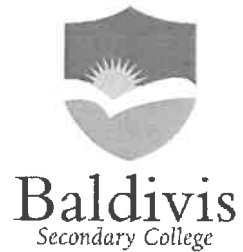
Solve the following pair of simultaneous equations graphically

$$\begin{cases} x + 4y = 8 \\ y = 9 - 2x \end{cases}$$



$$\begin{aligned} x &= 4 \\ y &= 1 \end{aligned}$$

Year 10
Pathway 10A



Term 3 Test
Statistics and Linear Relationships
Section Two

Score: /14
Total Time: 15 minutes

Name: _____

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks. Classpad calculators may be used. One A4 notes (handwritten) allowed.

Question 1

(6 marks)

Two classes of students sat the same test, one on Monday and one on Friday. Their results are shown below:

Monday: 14, 17, 18, 11, 20, 19, 20, 16, 13, 19, 4, 10, 15, 20, 18, 17, 18, 9, 14, 18, 13, 17.

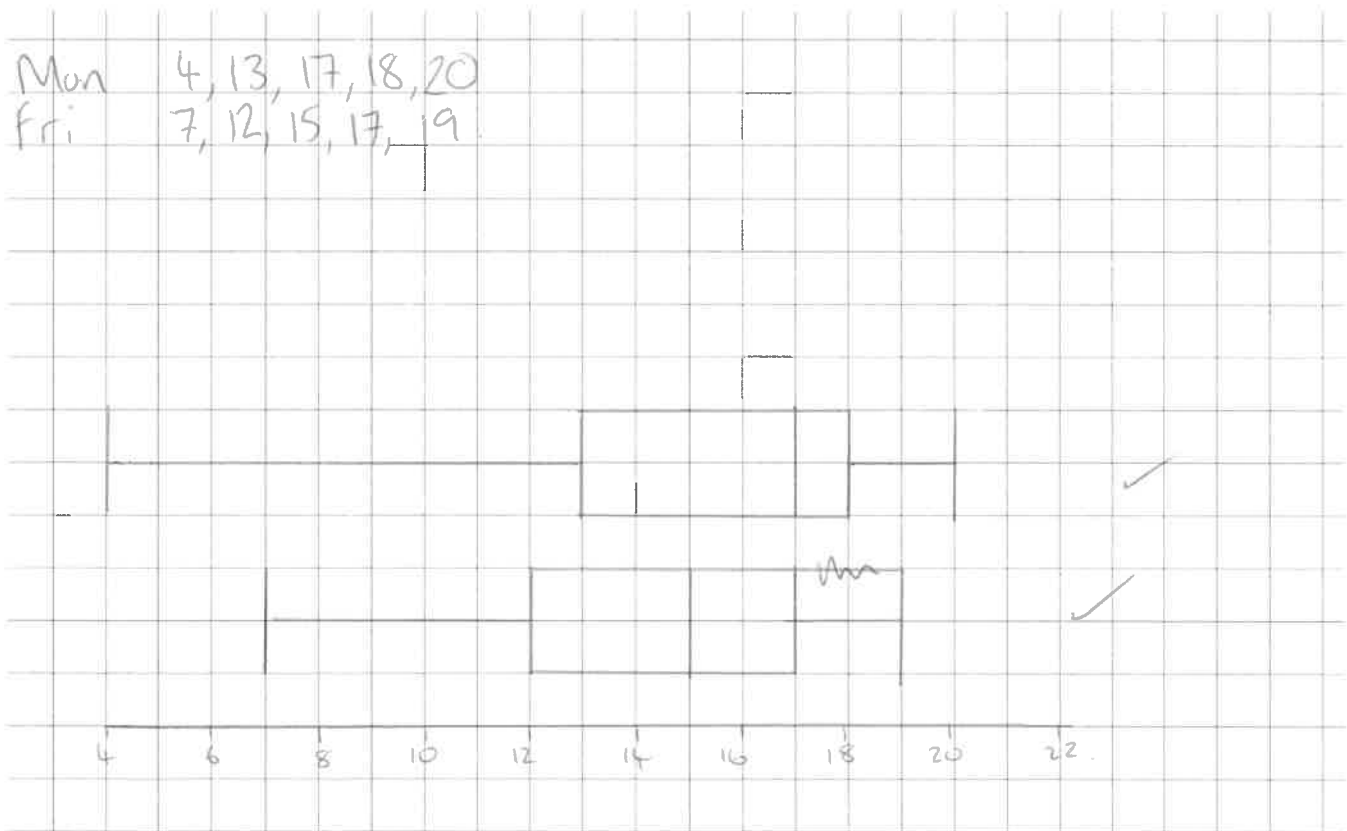
$$\bar{x} = 15.5$$
$$\sigma = 4.04$$

Friday: 19, 13, 16, 13, 11, 16, 19, 9, 7, 15, 17, 19, 16, 13, 9, 12, 16, 18, 15.

$$\bar{x} = 14.36$$
$$\sigma = 3.48$$

Compare the results for the two classes.

(Hint: Find the summary statistics and create the parallel box plots)



Comment - Measures of centre. —
- Measures of spread. —
- Skewness. —
- Relates to class results. —

Question 9

(7 marks)

Solve the following pairs of simultaneous equations

a) $\begin{cases} x + y = 22 \\ x = y + 8 \end{cases}$

$$\begin{aligned} 2y + 8 &= 22 \\ 2y &= 14 \\ y &= 7 \\ x &= 7 + 8 \\ x &= 15 \end{aligned}$$

b) $\begin{cases} 5x + 2y = 52 \\ x - 2y = 8 \end{cases}$

$$\begin{aligned} 6x &= 60 \\ x &= 10 \\ 10 - 2y &= 8 \\ y &= 1 \end{aligned}$$

c) $\begin{cases} 2x + 3y = 19 \\ 5x + 2y = 20 \end{cases}$

$$\begin{aligned} 10x + 15y &= 95 \\ 10x + 4y &= 40 \\ 11y &= 55 \\ y &= 5 \\ 2x + 15 &= 19 \\ 2x &= 4 \\ x &= 2 \end{aligned}$$

$$\begin{aligned} \text{or} \quad 4x + 6y &= 38 \\ 15x + 6y &= 60 \\ 11x &= 22 \\ x &= 2 \\ 4 + 3y &= 19 \\ 3y &= 15 \\ y &= 5 \end{aligned}$$

Question 10

(3 marks)

A group of 2 adults and 3 children paid a total of \$52.50 for their cinema tickets. Another group of 1 adults and 5 children paid \$59.50. What is the cost of an adult ticket and the cost of a child's ticket?

$$\begin{aligned} 2a + 3c &= 52.50 \\ a + 5c &= 59.50 \\ 2a + 10c &= 119 \\ 7c &= 66.50 \\ c &= \$9.50 \end{aligned}$$

$$\begin{aligned} 2a + 28.50 &= 52.50 \\ 2a &= 24 \\ a &= \$12 \\ \text{adult } \$12 \\ \text{child } \$9.50 \end{aligned}$$

Question 2

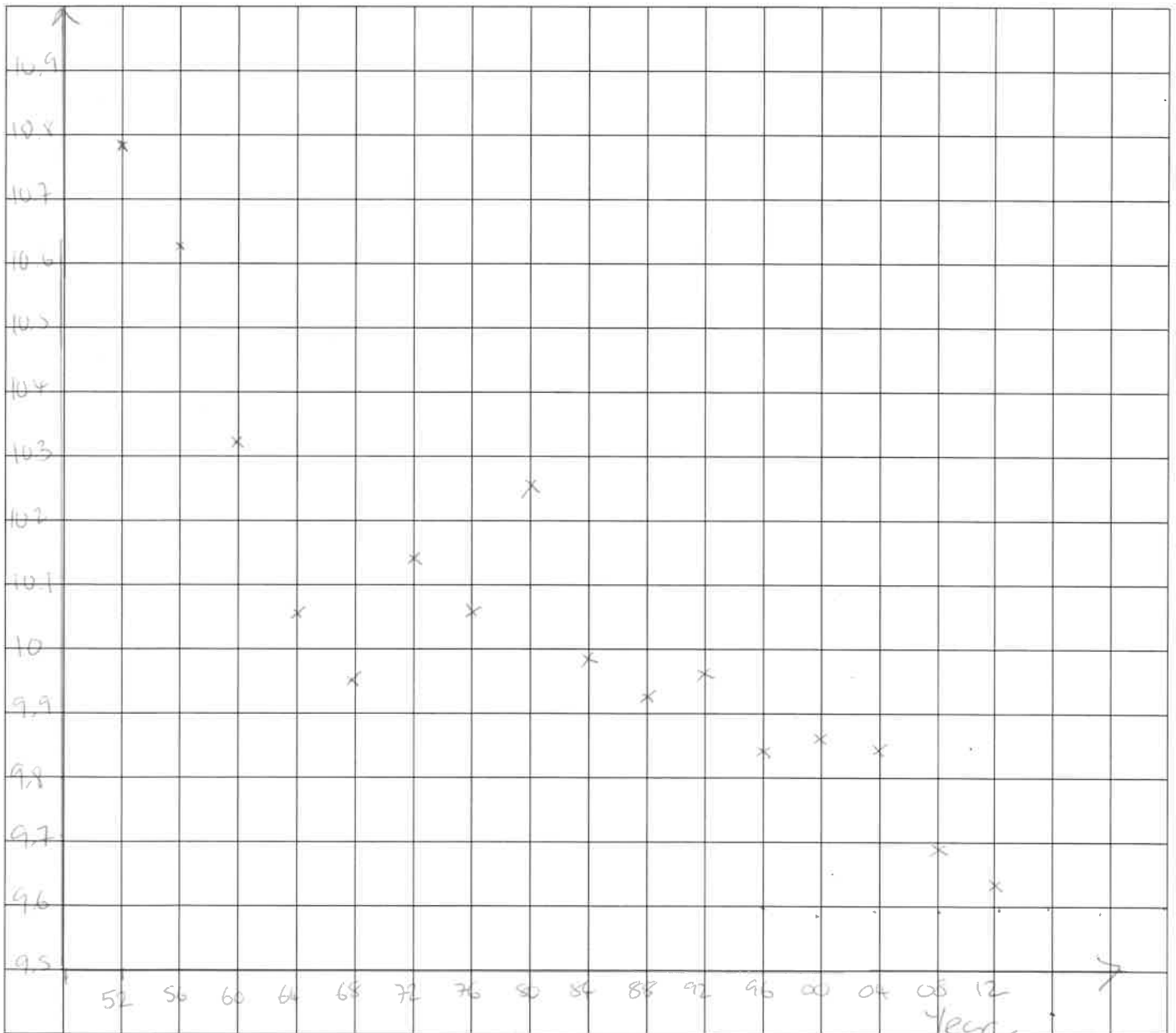
(8 marks)

Usain Bolt of Jamaica successfully defended his title and set a new race record of 9.63 seconds in the 100m sprint in the London 2012 Olympic Games. Consider this record of winning times for the 100m sprint since 1952.

Year	1952	1956	1960	1964	1968	1972	1976	1980
Time	10.79	10.62	10.32	10.06	9.95	10.14	10.06	10.25

Year	1984	1988	1992	1996	2000	2004	2008	2012
Time	9.99	9.92	9.96	9.84	9.87	9.85	9.69	9.63

a) Draw a scatterplot of the data.



b) Use your calculator to find the equation of the least squares regression line.

$$y = -0.014235x + 38.27$$

$$\Rightarrow y = -0.0142x + 38.3$$

c) In the 1960's, the speculation was 'When will the 10-second barrier be broken?' Today, talks centre around 'When will the 9.5-second barrier be broken?' Use your equation to make a prediction when this might occur.

2021.25 → In 2021 or next olympic games after this 2024
 Could give either with games being held 2021.

